MIL-C-43141D(ME)
15 November 1978
SUPERSEDING
MIL-C-43141C
17 January 1973

MILITARY SPECIFICATION

CUTTERS; PAPER, GUILLOTINE: FLOOR MODEL, ELECTRIC-MOTOR-DRIVEN

This specification is approved for use by all Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electric-motor-driven paper cutters.

1.2 Classification. Paper cutters covered by this specification shall be of the following sizes as specified (see 6.2).

Size 1	- 18-1/4-inch minimum throat width. 19-3/4-inch maximum throat width.
Size 2	- 23-inch minimum throat width. 26-1/2-inch maximum throat width.
Size 3	- 30-1/2-inch minimum throat width. 45-inch maximum throat width.
Size 7	- 46-1/2-inch minimum throat width. 57-inch maximum throat width.
Size 8	- 58-inch minimum throat width. 64-inch maximum throat width.
Size 9	- 65-inch minimum throat width. 89-inch maximum throat width.
Size 10	- 90-inch minimum throat width. 111-inch maximum throat width.

FSC 3610

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (for Treatment of Communications, Electronic, and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-P-3684	- Printing, Duplicating, and Book Binding Equipment, Packaging of.
MIL-G-3859	- Grease Gun, Hand Operated, Lever, Push and Screw Type.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-S-15769	- Straightedges, Steel (Machinists' and Toolmakers').

STANDARDS

MILITARY

MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.

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(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

SOUTHERN CALIFORNIA AIR POLLUTION CONTROL DISTRICT

Rule 102 - Photochemical Reactive Solvents.

(Application for copies should be addressed to the Southern California Air Pollution Control District, 434 South San Pedro Street, Los Angeles CA 90013.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Description. The paper cutters shall be motor driven and shall consist of a base and frame, table (air cushion, 45 inches and larger models), knife, knife-bar, cutting stick, paper clamp, back and side gages, safety guards, and a hydraulic system.

3.2 First article (first produced cutter). The contractor shall furnish one cutter for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail will produce cutters that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

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3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Safety. All rotating or reciprocating parts that are of such nature or so located as to become a hazard to operating personnel shall be fully enclosed or guarded. A two-hand interlock control requiring the simultaneous use of both hands to start the cutting stroke shall be provided. In addition, a built-in, nonrepeat device shall limit each operation to one stroke. An overload safety device shall be provided which shall limit or disengage the power when the knife strikes a tough foreign object, such as metal, on the downward stroke.

3.5 Capacities and dimensions. Capacities, dimensions, and weights of the paper cutters shall be as shown in Table I.

TABLE I. Capacities, dimensions, and weights.

Capacity, dimension, and weight	Size						
	1	2	3	7	8	9	10
Throat width							
Minimum (inches)	18-1/4	23	30-1/2	46-1/2	58	65	90
Maximum (inches)	19-3/4	26-1/2	45	57	64	89	111
Clearance under clamp (pile height)							
Maximum (inches)	2-1/2	3-1/2	5-1/2	7	7	7	8
Back gage from knife (limits of range):							
Far limit not less than (inches)	18-1/4	24	30-1/2	46-1/2	58	65	90
Near limit not more than (inch)	1/2	3/4	3/4	3/4	3/4	3/4	1
Back gage indication band graduations (inch)	1/16	1/16	1/16	1/16	1/16	1/16	1/16

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TABLE I. Capacities, dimensions, and weights. - Continued

Capacity, dimension and weight	Size						
	1	2	3	7	8	9	10
Front of table to knife:							
Minimum (inches)	12-1/4	18-1/2	18	21	19	21	25
Maximum (inches)	12-1/2	18-3/4	20	25	25	29	30
Table indication rule graduations (inch)	-	1/16	1/16	1/16	1/16	1/16	1/16
Width between size gages:							
Minimum (inches)	18-1/4	23	30-1/2	46-1/2	58	65	90
Maximum (inches)	19-3/4	26-1/2	45	57	64	89	111
Height of table top:							
Minimum (inches)	34	34	34	37-1/2	37	37	37
Maximum (inches)	34-1/2	36	37-1/2	39	40	40	40
Floor space:							
Width maximum (inches)	35	59	123	136	141	147	173
Depth maximum (inches)	45	63	98	122	128	150	173
Weight, maximum (pounds)	905	2,000	10,000	10,000	14,000	15,000	22,000

3.6 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings.

3.7 Accuracy. Corresponding edges of trimmed sheets shall not differ in length more than 0.03 inch, and adjacent edges of sheets shall be perpendicular within 5 minutes of arc. The trimmed surface of the pile shall be smooth and free of chatter marks, and the individual sheets shall have clean, straight edges.

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3.8 Base and frame. The base and frame shall be sturdy construction assuring a rigid and firm structure for the operating parts of the machine.

3.9 Table. The table shall have a top surface lying in a plane with an overall variation of not more than plus or minus 0.005 inch, and variation in any 12-inch distance shall be not more than plus or minus 0.003 inch. The table surface in front of the knife shall be provided with indication rule graduations. A recess shall be provided for holding the cutting stick (see 3.12) and shall extend over the full width of the table beneath the knife edge. The recess shall be slightly off center from the cutting edge of the knife to provide cutting surfaces through rotation and reversal of the stick. The working area shall be provided with uniform illumination and when specified (see 6.2) include a light line indicator.

3.10 Knife. The cutting knife shall be steel best suited for this type of equipment. The knife shall be designed to provide solid seating and bolting to the knife bar. Not less than two separate settings shall be provided on the knife to compensate for wear. Each cutter shall be equipped with six cutting knives and storage safety guards.

3.11 Knife bar. The knife bar shall have a machined seat for solidly seating the knife. Adjusting and leveling devices shall be provided to insure correct contact of the knife blade with the cutting stick when the bottom of the cutting stroke has been reached. The face and back of the knife bar shall have machine-finished bearing surfaces.

3.12 Cutting stick. The cutting stick shall be not less than 0.5 inch or more than 0.75 inch square and shall fit snugly within the recess provided in the table. Each cutter shall be equipped with 36 cutting sticks.

3.13 Paper clamp. A manually-controlled, hydraulically-operated clamp shall be provided to firmly hold the paper stock during the cutting operation. Adjustment of the clamp pressure on the stock and the amount of pressure shall be indicated on a gage. The clamp shall be provided with a detachable sole plate of sufficient length to cover the width of the clamping surface.

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3.14 Gages.

3.14.1 Back gages.

3.14.1.1 Power model. The back gage may be manually operated or power driven. The gage shall maintain the setting when moved to any distance from the knife within the range specified in Table I. The distance between the gage and the knife shall be measured on a corrosion-resistant steel band with an accuracy of 0.01 inch. A centrally located dial, illuminated and magnified, shall be built into the cutter for reading the band from the operator's position. A locking device, with a control located at the front of the cutter, shall be provided to secure the setting against movement during the cutting operation. The face of the gage shall fit the surface of the table in such a manner that any movement of the gage forward shall guide the complete pile of paper stock.

3.14.1.2 Spacer model. The back gage shall be motor driven with a built-in electric brake on the lead screw to positively lock the back gage in place. Positioning of the back gage shall be controlled by a magnetic signal. Magnetic signals shall be placed on a magnetic tape or cylinder or erased by remote switches to provide for programs of "forward" stop positions and "reverse" limits to travel on any of 16 to 24 channels. Any combination of channels shall be usable for programs. Programs shall be set up by (a) on machines using the tape - positioning the back gage where stops are desired and marking the tape at those positions, or (b) on machines using the cylinder - while cutting the first lift of stock mark the stop positions on the cylinder by the "cut-mark" method. The back gage shall start forward automatically after each cutting cycle of the knife. On cutters over 37 inches a two-speed drive system, both fast and slow, shall control the lead screw. Both speeds shall be used for forward. Fast to be used for reverse only. The slow speed shall be used to decelerate the back gage before stopping at a stop mark. The back gage may be operated independently from the spacer mechanism by the fast, slow, manual control buttons.

3.14.2 Side gages. The right- and left-side gages (guide plate) shall be fixed perpendicular to the horizontal plane of the table and perpendicular to the vertical plane traveled by the knife.

3.15 Electrical requirements. The cutters shall operate on the electric power specified (see 6.2). All electrical equipment and wiring shall conform to NFPA No. 70.

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3.15.1 Motors. The electric motors shall conform to CC-M-1807, Type II, Class 3, Style E, Enclosure 1, Design A.

3.16 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, Composition I or II, as applicable (see 3.16.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.16.1 Composition. Compostion II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of Composition I. When Composition II is used, the contractor shall provide evidence to the Government that the use of Composition II is required, and shall certify that the Composition II material complies with Rule 102, Southern California Air Pollution Control District.

3.17 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.17.1 Lubricants. The cutters shall operate as specified herein when lubricated with military lubricants conforming to MIL-G-10924 and MIL-L-2104.

3.17.2 Lubrication fittings. Lubrication fittings shall conform to MS15003. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

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3.17.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.18 Identification marking. The cutters shall be identified in accordance with MIL-STD-130. The marking shall be applied to the cutters on identification plates conforming to MIL-P-514, Type I, Style I, Composition C of Type I, Grade A, Class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.19 Instruction and warning plates. Each cutter shall be equipped with instruction plates or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the cutter. Plates shall conform to MIL-P-514, Type III, Composition C of Type I, Grade A, Class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.20 Treatment and painting. The portions of the cutter normally painted shall be cleaned, treated, and painted in accordance with the contractor's usual standard practice.

3.21 Workmanship. All parts, components, and assemblies of the cutters including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or beveled except where sharp edges are required.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced cutter inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced cutter inspection.

4.3.1 Examination. The first produced cutter shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced cutter shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each cutter shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each cutter shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The cutters shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Safety requirements not as specified.
103. Capacities and dimensions not as specified.
104. Maintainability features not as specified.
105. Parts, components, and accessories not as specified or missing.
106. Fungus and moisture resistance treatment not as specified.
107. Lubrication not as specified.
108. Identification marking missing, incomplete, or illegible.
109. Instruction and warning plates not as specified or missing.
110. Workmanship not as specified.

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4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, the cutters shall be lubricated with oils and greases conforming to MIL-L-2104 and MIL-G-10924. Oils and greases shall be those designated for use in the ambient temperature at which the test is being conducted.

4.5.2.2 Table flatness. The flatness of the table surface shall be determined using a straightedge conforming to MIL-S-15769 and a feeler gage. Variation in excess of that specified in 3.9 when checking the table surface parallel to the edges and the diagonals shall constitute failure of this test.

4.5.2.3 Accuracy.

4.5.2.3.1 Squareness and length of sheet. A pile of 24-pound book paper of the height specified in Table II shall be trimmed to the dimensions specified in the table. Sample sheets shall be selected from near the top and bottom of the cut pile and examined. Nonconformance to 3.7 shall constitute failure of this test.

TABLE II. Height of pile and dimensions.

Cutter size	Height of pile Inches	Trimmed dimensions Inches
1	2	1/2 x 17 8-1/2 x 11 11 x 17
2	3	3/4 x 25 17 x 22 22 x 25
3	5	3/4 x 29-1/2 10 x 12 34-1/2 x 44-1/2
7	6-1/2	3/4 x 46 35 x 45 46 x 46
8	6-1/2	3/4 x 56 23 x 36 56 x 56

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TABLE II. Height of pile and dimensions. - Continued

Cutter size	Height of pile Inches	Trimmed dimensions Inches
9	6-1/2	3/4 x 63 23 x 36 63 x 63
10	7-1/2	1 x 89 34 x 44 89 x 89

4.5.2.3.2 Indicator band reading. At completion of the cuts required in 4.5.2.3.1, the indicator band reading shall be checked. Nonconformance to 3.14.1 shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking of the cutters for Level A or Level B shall be examined for conformance to the quality assurance provisions of MIL-P-3684. The preservation, packing, and marking for Commercial shall be examined to determine compliance with MIL-STD-1188.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.1.1 Level A. Preservation of the cutters shall be in accordance with Level A requirements of MIL-P-3684.

5.1.2 Level B. Preservation of the cutters shall be in accordance with Level B requirements of MIL-P-3684.

5.1.3 Commercial. Preservation of the cutters shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Packing shall be in accordance with the Level A requirements of MIL-P-3684.

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5.2.2 Level B. Packing shall be in accordance with the Level B requirements of MIL-P-3684.

5.2.3 Commercial. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military packaging shall be in accordance with MIL-P-3684.

5.3.2 Commercial. Marking for Commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The cutters are intended to be used for cutting, squaring, and trimming piles of paper for printing of maps, forms, and miscellaneous documents in both fixed site and mobile van environments.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Time frame required for submission of first produced cutter (see 3.2).
(d) Light line indicator required (see 3.9).
(e) Electric power required (see 3.15).
(f) Degree of preservation and degree of packing required (see 5.1 and 5.2).

6.3 First produced cutter. Any changes or deviations of production cutters from the approved first produced cutter during production will be subject to the approval of the contracting officer. Approval of the first produced cutter will not relieve the contractor of his obligation to furnish cutters conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each cutter.

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6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of cutters.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodian:
Army - ME

Review activities:
Army - GL
DLA - GS

Preparing activity:
Army - ME

Project 3610-A302

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DOCUMENT IDENTIFIER AND TITLE: MIL-C-43141D(ME) Cutters; Paper, Guillotine: Floor Model, Electric-Motor-Driven

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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